Nicholas Eisenberg, Ph.D.

702-354-0866 || nickeisenberg@gmail.com || linkedin.com/in/nick-eisenberg || github.com/nickeisenberg

Experience

Senior Data Scientist *June 2022 - Present*
Nevada National Security Site

- Designed an anomaly-detection model in PyTorch using an LSTM variational-autoencoder to detect component failure in a pulse-power x-ray machine. The model was deployed to the machine operators with Django and Docker and cut machine repair times by 50%
- Applied the U-Net architecture with PyTorch to develop a deblurring model for a pulse-powered x-ray machine. The model was applied by the machine operators and reduced machine configuration time by 30%.
- Utilized YOLO v5 to develop an object-detection model in PyTorch that achieved a 80% mAP evaluation on an in-house thermal-image vehicle dataset for a R&D project.
- Designed an A/B hypothesis-based anomaly detection model using Statsmodels for a radiation detection R&D project, utilizing data from continuously streaming sensors.
- Enhanced the scalability of models initially prototyped by our team of Analysts and Data Scientists for training on our internal high-performance computing (HPC) cluster. My role involved refactoring the existing code to enable efficient execution in a distributed and parallel setting with PyTorch's DistributedDataParallel.
- Developed and deployed a Django web interface with Docker Compose, enabling non-technical staff to add nodes and connections to the company's Neo4j graph database, without knowledge of the query syntax.

Personal Projects

iron.nvim Maintainer (GitHub link)
iron.nvim is an interactive and cross operating system REPL plugin for neovim, written in Lua, that supports many interpretted languages such as Python, R, Lua and more. I have joined the developers of iron.nvim to help maintain the code base, add new features and respond to ongoing issues.

Technical Skills

Programming Languages: Python, SQL, Lua, Bash, Cypher, LaTeX
Libraries: PyTorch, Pandas, Scikit-Learn, Sqlalchemy, Matplotlib, Plotly, Django, Streamlit
Developer Tools: Git, AWS, Unix/Linux, Vim, Tmux, SLURM, Docker, Neo4j

Education

Auburn University - Ph.D. *June 2022*
Research Area: Probability (Stochastic Processes and Partial Differential Equations)
Advisor: Le Chen

University of Nevada, Las Vegas - B.S./M.S. *December 2015 / January 2018*
Major: Applied Mathematics

Publications

1: Eisenberg, N., Chen, L. Interpolating the stochastic heat and wave equations with time-independent noise: solvability and exact asymptotics. Stoch PDE: Anal Comp (2022).
https://doi.org/10.1007/s40072-022-00258-6

2: Eisenberg, N., Chen, L. Invariant Measures for the Nonlinear Stochastic Heat Equation with No Drift Term. J Theor Probab (2024). https://doi.org/10.1007/s10959-023-01302-4